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Comparative analysis of YOLO models for green coffee bean detection and defect classification

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The quality and uniformity of green coffee beans significantly influence the overall flavor and value of the product. In the coffee industry, automated flaws and bean-type identification offer numerous advantages. This study examines the effectiveness of multiple YOLO (You Only Look Once) models for identifying and classifying green coffee beans. Various YOLO variants, including YOLOv3, YOLOv4, YOLOv5, YOLOv7, YOLOv8, and custom models, are compared with a focus on computational efficiency, accuracy, and speed. Utilizing a dataset of 4,032 training and 506 testing images encompassing diverse bean types, defects, and lighting conditions, we assessed the performance. The bounding boxes generated by our models accurately encompass coffee beans, with the background typically uniform and set to black. Our analysis reveals the superior performance of the custom-YOLOv8n model, which we meticulously customized for green coffee bean detection. This model achieved high precision, recall, f1-score, and mAP, demonstrating its potential for real-world implementation in coffee bean quality control systems. The customization process involved fine-tuning the model to focus on significant features relevant to green coffee bean detection and employing specific labeling strategies. Customization allows you to fine-tune the model to focus on important features relevant to green coffee bean detection. This sensitivity ensures that the model can effectively distinguish between different bean types and detect even subtle defects. This paper clarifies our primary objective of evaluating YOLO models' performance in identifying and categorizing green coffee beans, with potential implications for enhancing efficiency and consistency in the coffee industry. A succinct key sentence underscores the benefits of our research for readers seeking efficient YOLO model selection and implementation in agricultural systems.

Keywords Analysis, Classification, Coffee bean, Custom-YOLOv8n, Defect, Detection, and Evaluation metrics

The quality control of green coffee beans plays a crucial role in determining the flavor and commercial value of the final product. In the coffee industry, automating the detection and classification of green coffee beans can significantly enhance productivity, reduce labor costs, and ensure consistent quality in production. Traditionally, manual inspection has been the dominant method for sorting green coffee beans in many coffee-producing countries. This manual process, however, is time-consuming, labor-intensive, and prone to human error, leading to inconsistencies in quality assessment[1]. As such, there is a pressing need for an automated solution capable of providing fast, accurate, and reliable classification of green coffee beans and defects.

Recent advances in deep learning, particularly in object detection models like YOLO (You Only Look Once), have revolutionized image-based processing tasks in agriculture and food processing. YOLO models offer real-time object detection with high accuracy and efficiency, making them highly applicable to tasks like agricultural inspection, where rapid and accurate classification is essential[2,3]. For example, previous studies have successfully applied YOLO models in the detection of apple blossoms[4], tomatoes[5], cherries[6], and apple[7], yielding excellent performance in detecting small objects with high precision. Similarly, the use of YOLOv5 has been explored in

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the detection of apples by harvesting robots[8], as well as in poisonous mushroom identification[9], where YOLO's speed and accuracy proved advantageous.

Despite the success of YOLO in various agricultural applications, the unique morphology and size heterogeneity of green coffee beans present specific challenges. The variation in bean size, shape, and surface defects necessitates robust model customization to ensure detection accuracy. In this study, we undertake a comprehensive comparative analysis of multiple YOLO models, including YOLOv3, YOLOv4, YOLOv5, YOLOv7, and YOLOv8, as well as custom models specifically tailored for green coffee bean detection and defect classification.

Our dataset consists of 4,032 training and 506 testing images, capturing different types of beans, defects, and lighting conditions. We standardized image resolution to 640×640 pixels to maintain consistency across the dataset, allowing YOLO's multi-scale detection capabilities to handle the inherent size variability of coffee beans. Through this approach, we aim to address a major bottleneck in existing methods-how to reliably detect heterogeneous objects with significant size and morphological variations[10,11].

The results of our study show that the custom YOLOv8 model outperforms the other variants in terms of precision, achieving a mean Average Precision (mAP) of 0.995 and a precision of 0.987. The model was fine-tuned by modifying key components such as the backbone, neck, and head, along with optimizing hyperparameters like the learning rate and momentum. These enhancements enabled the model to focus on the relevant features of green coffee beans, improving both defect detection and classification accuracy. The proposed model offers significant advantages, including improved detection of small defects, reduced processing time, and increased reliability, making it highly suitable for real-world coffee bean inspection applications[12,13].

By automating the detection of defective green coffee beans, our model contributes to enhanced quality control in the coffee industry. Moreover, this research provides a foundation for future innovations in agricultural automation, paving the way for more efficient and scalable solutions in food production systems. The findings presented in this paper have important implications for coffee producers and processors, offering a practical tool for improving product quality and consistency in a highly competitive global market.

The outline of the paper is organized as follows. Section "Materials and methods" introduces the materials of green coffee beans and methods. In Section "Results and discussion", the experimental results are presented with a detailed discussion of comparative analysis. Finally, Section "Conclusions and future works" concludes and draws future work.

Materials and methods

Data description

There are four types (such as black, broken, fade and sour) of data in the green coffee bean dataset. Green coffee beans collected from coffee farmers in Timor-Leste have been sorted by hand. Coffee beans were placed on A4-size white paper without touching each other. The green coffee bean images were taken using a Nikon (D5100) digital camera (F/16 f-number, ISO 200, exposure compensation of 1.3, exposure time of 1/60 s, autofocus mode). The camera is mounted 45 cm above the target object using compatible steel. Three lighting devices were placed to increase the brightness around the coffee beans, as shown in Fig. 1[14–16]. The green coffee bean images were taken at the maximum resolution of the digital camera. The images were separated into each bean by a program using OpenCV (an image processing library for Python). The normal green coffee beans are notably distinct from the various types of defective coffee beans (as illustrated in Fig. 2).

The images are all in JPG format. The image annotation files contain 5044 images. The images are all in JPG format. As a preprocessing step, all images were scaled to the same size and the image size was set to 640×640 pixels. These files specify the precise locations of the objects with labels in the corresponding image. The annotation was done manually, and the annotated values were also saved in txt files. The data files were divided into two folders: images and labels. The green coffee bean dataset is split into 80% for training, 10% for validation, and 10% for testing. The training, validation, and test dataset used 4032, 506, and 506 green coffee bean images, respectively, which are shown in Table 1. It should be noted that one coffee bean image may contain multiple defective labels.

The images were resized to a standard resolution of 640×640 pixels for all models to maintain consistency. As a preprocessing step, all images were scaled to the same size. The images were manually labeled as different types of green coffee beans. These data classify coffee beans into four labels (black, broken, fade, and sour) as shown in the Fig. 2b. It should be noted that one coffee bean image might contain multiple labels. All images of the coffee beans were divided into three groups: training data (80%), validation data (10%), and test data (10%) (as shown in Table 1).

Data augmentation techniques, such as random rotations, flips, and brightness adjustments, were applied to the training dataset to improve model robustness and reduce overfitting. For the architecture of the custom-YOLOv8n model, it is recommended that more detailed architectural diagrams and explanations be provided so that other researchers can reproduce and understand the design of the model.

Types of coffee beans

Black bean: Unroasted beans are black or very dark beans. Black beans typically result from harvesting immature cherries or by harvesting dead cherries that fall naturally from the tree[1,16–18].

Broken bean: Broken beans refer to wet-processed beans that have been cut or bruised during the pulping process[1,16–18].

Fade bean: Fade beans are unroasted coffee beans that have lost a large part of their original color, a characteristic of old crops and beans that dried too rapidly[1,16–18].

Sour bean: Sour can be caused by overripe or fermented coffee beans or improper fermentation[1,16–18].

Fig. 1. The figure shows the experimental setup[14–16].

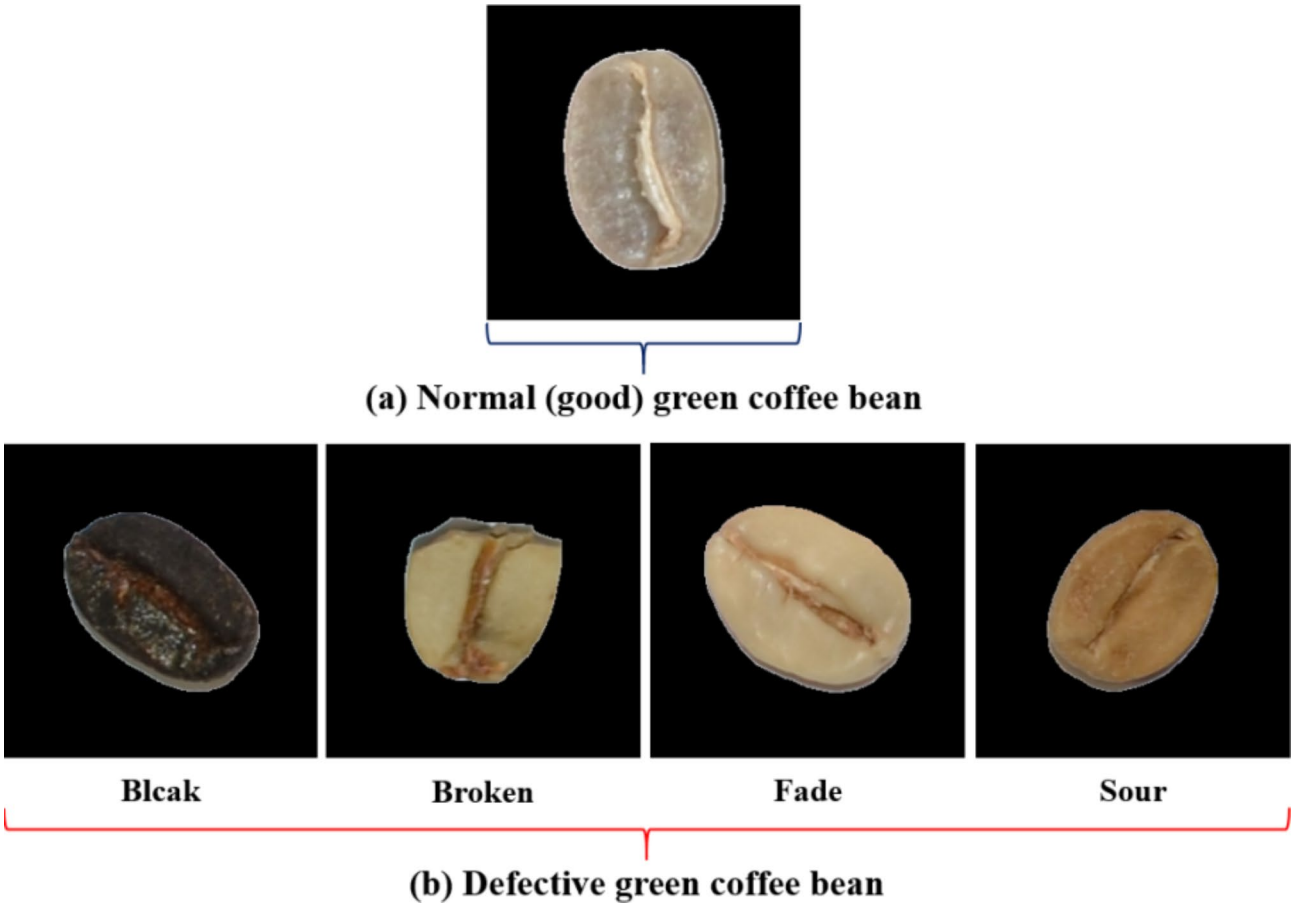

Fig. 2. The figure shows the different types of green coffee bean images.

Name	Training	Validation	Test	Total
Coffee bean	4032	506	506	5044

Table 1. Defective green coffee bean image distribution.

Data pre-processing and annotation

After taking various defective images of green coffee beans, the images were all converted to JPG format. Popular annotation tool Roboflow was used to cautiously label the images in this final phase. First, each image is opened in this tool one at a time. Then, a rectangular shape was manually drawn to the boundary of an object to specify its exact location in that image by x_{center}, y_{center}, width and height. Finally, each object has been given a label, such as 'black', 'broken', 'fade' and 'sour'. In Roboflow, annotated values were saved as txt files in YOLO format.

Even while the color and texture of sour beans may resemble those of regular beans to some extent, they can nevertheless have subtle yet distinct characteristics like:

(a) Surface distortion or wrinkled texture.
(b) Slightly different hue (more reddish or discolored, as shown in our dataset).
(c) Variations in gloss or shine that our system picks up on.

Although these characteristics aren't usually very noticeable to the naked eye, deep learning models like YOLO, which can recognize these subtle patterns across enormous datasets, can identify them with great accuracy. Due to the nuanced nature of these faults, traditional techniques of diagnosing sour beans primarily rely on manual inspection, which is time-consuming and prone to human error. This is eliminated by our strategy by:

Offering dependable detection based on a vast array of training examples, which increases its dependability over arbitrary human examination. Saving money on labor and time because the YOLO model processes images far more quickly than a human can.

To comply with the fixed input size requirement of the YOLO models, all input photos in our investigation were scaled to a uniform resolution of 640 × 640 pixels. In order to guarantee uniformity throughout the dataset and provide a standardized input format for the detection and classification procedure, this scaling phase was added. Even with this consistent enlargement, the beans' aspect ratio was kept constant to prevent distortion and retain the original dimensions of every bean. Furthermore, the model can adjust to items of varied sizes inside the image, including coffee beans with differing morphologies, thanks to YOLO's built-in multi-scale detection capability. Furthermore, during training, data augmentation methods like scaling, rotation, and cropping were used to improve the model's capacity to generalize across various bean sizes and shapes. By taking these precautions, the model is able to detect and categorize beans of various diameters with accuracy, as the heterogeneity in the size and morphology of green coffee beans is successfully managed.

Bounding box

The bounding boxes data in YOLO is formatted as $[class,\ x_{center},\ y_{center},\ width,\ height]$ .The difference can be seen in Fig. 3. With the minimum point $(x_{min},\ y_{min})$, the center point $(x_{center},\ y_{center})$ can be calculated as below:

$$x_{center} = x_{min} + \frac{width}{2}$$
$$y_{center} = y_{min} + \frac{height}{2}$$

The top-left corner's coordinates are indicated by xmin and ymin in the bounding box format, while the normalized center coordinates, scaled between 0 and 1, correspond to the image's dimensions (as shown in Fig. 3).

Importing the dataset

The green coffee bean dataset was downloaded to Collab using the Roboflow-generated URL as a zip folder. The zip folder has been unzipped and deleted.

Experimental setup

The model was trained using the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.01 and a weight decay of 0.0005, 0.001. It was trained for 150 epochs with a batch size of 16. We kept the training process consistent for all the models and only changed the backbone. The training was performed on a single NVIDIA Tesla T4 GPU with 16 GB of memory. We used several libraries for our experiment, including PyTorch, OpenCV, Albumentations, Matplotlib, and Seaborn.

Detection of defective coffee bean

The coffee bean detection algorithm was implemented using open-source neural structures written in Python. Identification and classification were made using an object detection system called You Only Look Once (YOLO). This particular system is famous for its high processing speed. It processes images and predicts the possibility of belonging to a single step, box bounding object and class in a single step[19].

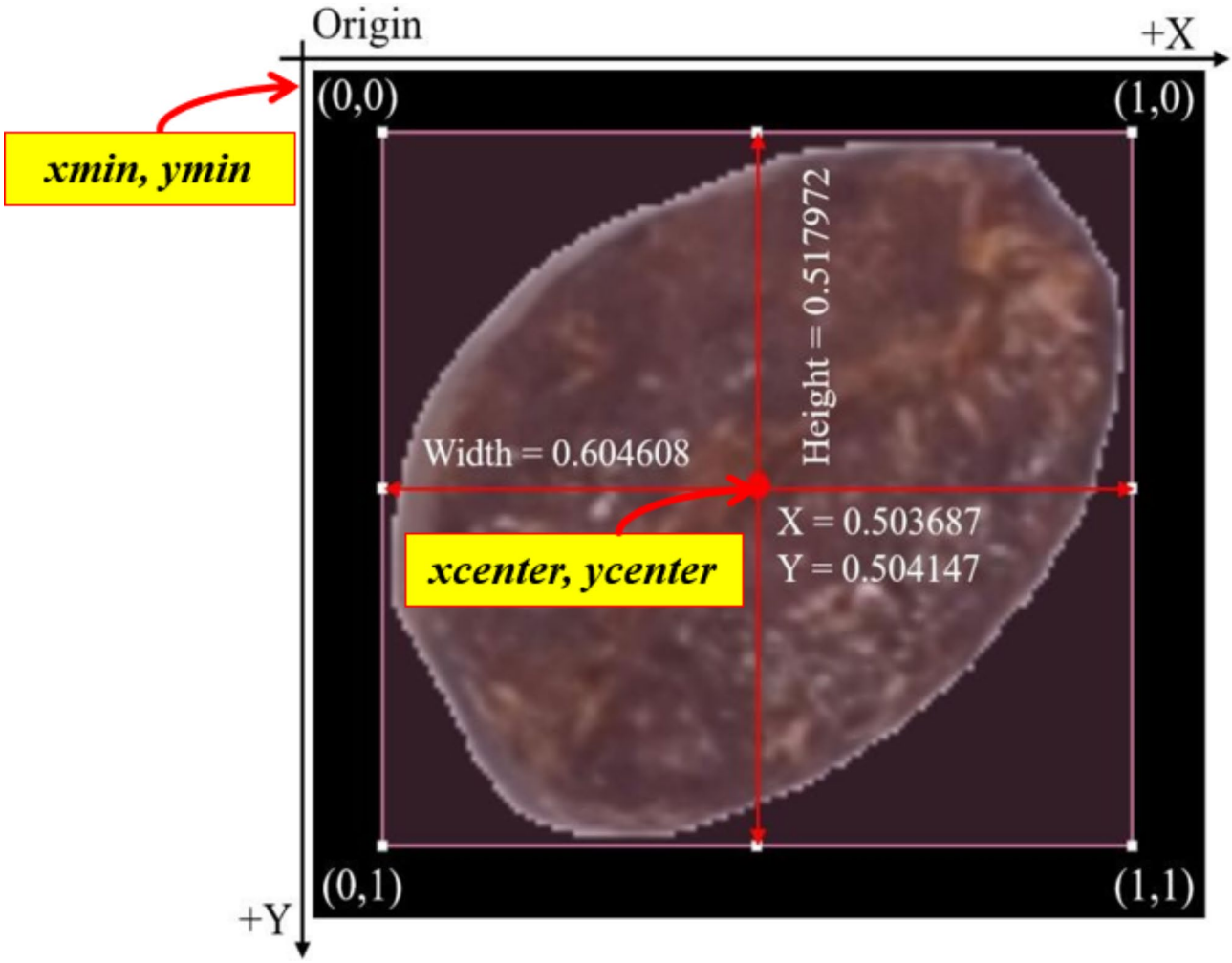

Fig. 3. The bounding box is in YOLO format. YOLO predicts the width and height of the box.

Research methodology

This diagram (Fig. 4) shows the process of training and evaluating a custom-YOLOv8n model using a dataset, split into training, validation, and testing sets.

Here's the detailed explanation:

Training Phase:

1. Dataset: The complete dataset is collected and prepared for the entire process. This dataset contains labeled images that will be split into training, validation, and testing sets.
2. Training Set: A portion of the dataset is assigned as the training set, where the model learns the characteristics of the data through iterative learning.
3. Extended Training Set: In some cases, data augmentation or additional data is added to the training set to improve the model's performance and ensure it generalizes well across different situations.
4. Custom-YOLOv8n: This refers to the specific YOLO (You Only Look Once) version 8 model that has been customized for this project. YOLO is a popular real-time object detection model that predicts bounding boxes and class labels in one pass through the network.
5. Our Model: This represents the model after it has been trained using the YOLOv8n architecture. The trained model has learned how to detect and classify objects from the training data.

Validation Phase:

6. Validation Set: During training, the validation set (another subset of the dataset) is used to tune the hyperparameters and prevent overfitting. The validation set helps in monitoring the model's performance on unseen data without directly adjusting weights based on the validation data.

Testing Phase:

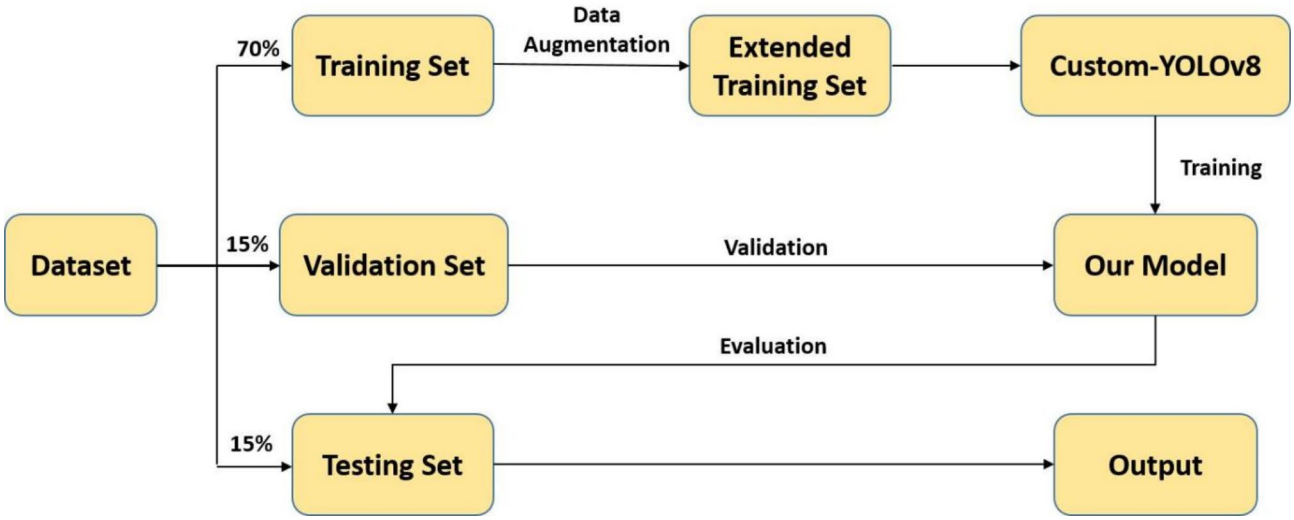

Fig. 4. Research methodology.

7. Testing Set: Once the model has been trained, it is evaluated on a completely separate set of data that the model has never seen before the testing set. This helps to assess how well the model generalizes to new data.
8. Output: The final output of the model, which can be evaluated using metrics such as precision, recall, or other performance measures relevant to the object detection task.

In summary, the diagram illustrates the step-by-step process of training a custom YOLOv8n model using a dataset that is split into training, validation, and testing sets. It emphasizes how the model is trained and refined using extended datasets and then evaluated for its performance on unseen test data.

The custom-YOLOv8n model

The provided image represents the architecture of a customized YOLOv8n model specifically designed for detecting defects in coffee beans. Customization of object detection models like YOLOv8n can enhance their sensitivity to specific defects in coffee beans by incorporating domain-specific knowledge into the model's architecture and training data. The backbone of the YOLOv8n model is altered by replacing the C3 module with C2 blocks based on the CSP concept. This modification likely aims to improve feature extraction for the particular characteristics of coffee bean defects. The C2f module, which combines C3 with ELAN, is introduced in the backbone. This is likely inspired by the ELAN concept in YOLOv8n, and it further enhances feature extraction for defect recognition. The architecture is divided into two main components: the backbone and the head. The backbone, which consists of several convolutional layers and C2 blocks, which is based on the CSP concept, replaces the C3 module in customv8, while the Backbone portion remains essentially the same as in YOLOv5 and is responsible for extracting hierarchical features from the input image of a coffee bean. The C2f module combines C3 with ELAN, taking inspiration from the ELAN concept in YOLOv7.These features are then passed to the head of the model, which includes additional convolutional layers and C2F blocks for further processing. The head also incorporates upsampling and concatenation operations to combine low-level and high-level features, enhancing the model's detection capability. This fusion of multi-scale features ensures that defects of various sizes and characteristics can be effectively identified. The Neck module was created by the YOLOv8n creators using two upsampling, and many C2f modules, The most widely used SPPF module was still utilized at the end of the backbone, and three 5×5 MaxPools were sent in serial before each layer was concatenated to ensure both lightweight and target accuracy across a range of scales. PAN-FPN feature fusion, which enhances the fusion and usage of feature layer information at various scales, is still employed by YOLOv8n in the Neck section, and the final decoupled head structure. YOLOv8n applied the concept of detaching the head from the neck to the last portion of the neck. It attained a new degree of accuracy by combining regression boxes with confidence.

The final detection is made in the detect block, where bounding boxes along with confidence scores are assigned to localize and quantify the detected defects on the coffee beans. This architecture ensures a comprehensive analysis by capturing intricate details through its multi-scale feature integration mechanism. The custom model is compatible with all YOLO versions and allows for version switching at any time. Its remarkable versatility also extends to running on different hardware platforms (CPU-GPU). The custom network architecture design is seen in Fig. 5.

The Training Phase and the Testing Phase are the two primary stages of the process of constructing a YOLO model for object identification that is depicted in the diagram. Gathering a variety of photographs, pre-processing them, and putting together a training set for the YOLO model are all part of the training phase. In the testing phase, a fresh picture is presented and the trained neural network model processes it to locate and identify items (as shown in Fig. 6). The YOLOv8n model, with its multi-scale feature integration mechanism,

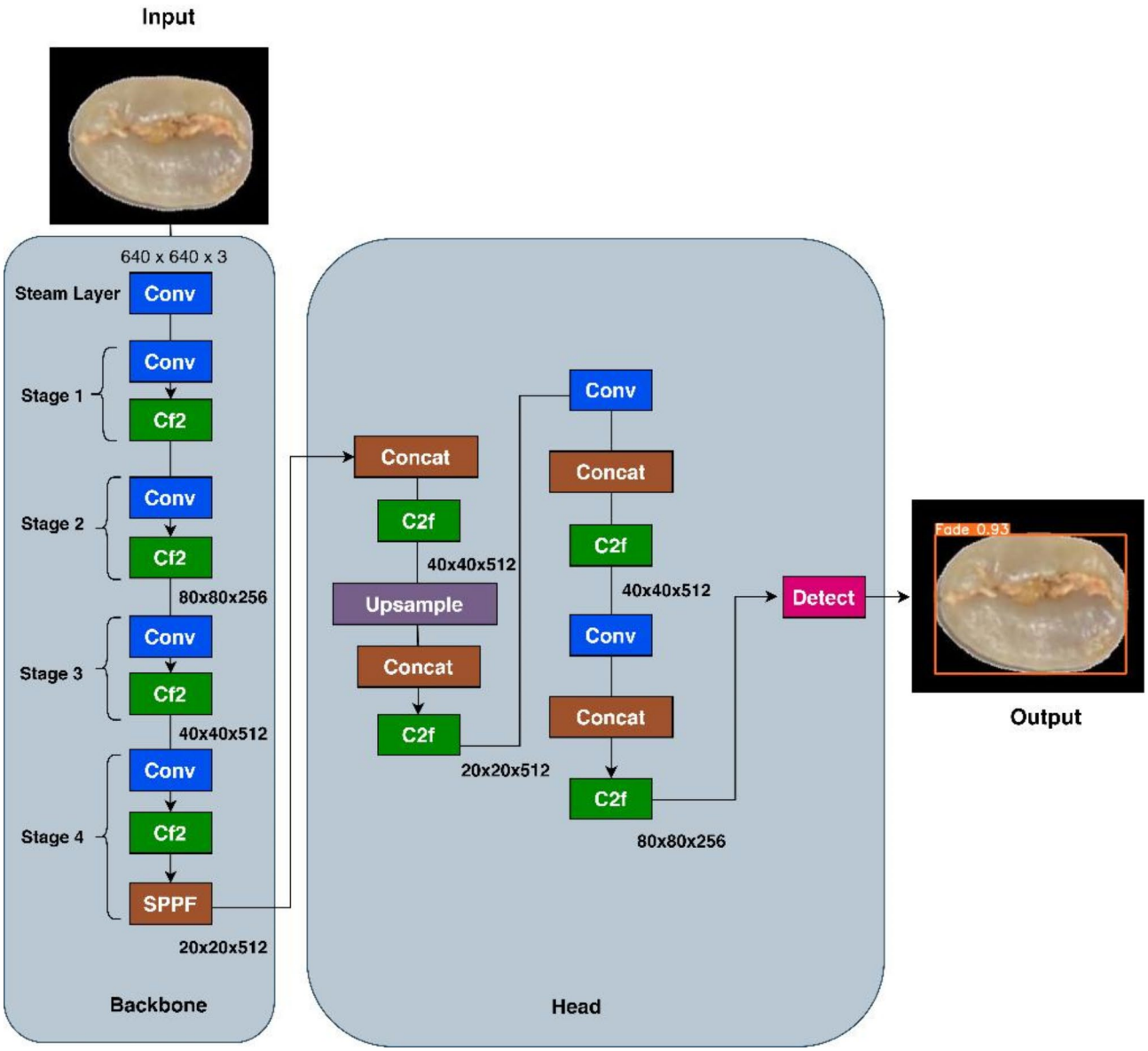

Fig. 5. The architecture of the customized YOLOv8n model.

is optimized for capturing intricate details of coffee bean defects, enhancing detection accuracy, and enhancing network performance. To provide a fair and precise detection method, the graphic also contains a donut chart that shows a balanced approach to training the model across several categories.

Experimental configuration

In this study, the YOLOv8 model was downloaded from the website https://github.com/ultralytics/ultralytics. The latest YOLOv8 release model includes four models of different sizes: YOLOv8n, YOLOv8s, YOLOv5m, YOLOv8l, and YOLOv8x[20]. In Our work YOLOv8n was selected in this study due to the compromise of its modest size, speed and outstanding performance. The network was pre-trained with the COCO dataset, and it was fine-tuned using the defective coffee bean dataset mentioned above (Table 1)[21]. Table 2 represents the configuration of the hyper-parameter.

Evaluation metrics

This study calculates precision, recall, and f1-score to evaluate the model performance. The evaluation indicators are computed based on the number of true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP) in an example (i.e., an image with multiple labels)[22]. Precision measures the correct predictions when the predicted bounding boxes match the ground truth boxes. On the other hand, recall measures the probability of accurate detection of ground truth objects. F1-score describes the combined measure of recall and precision considering their harmonic mean, where an f1-score of 1 represents the best value. The precision,

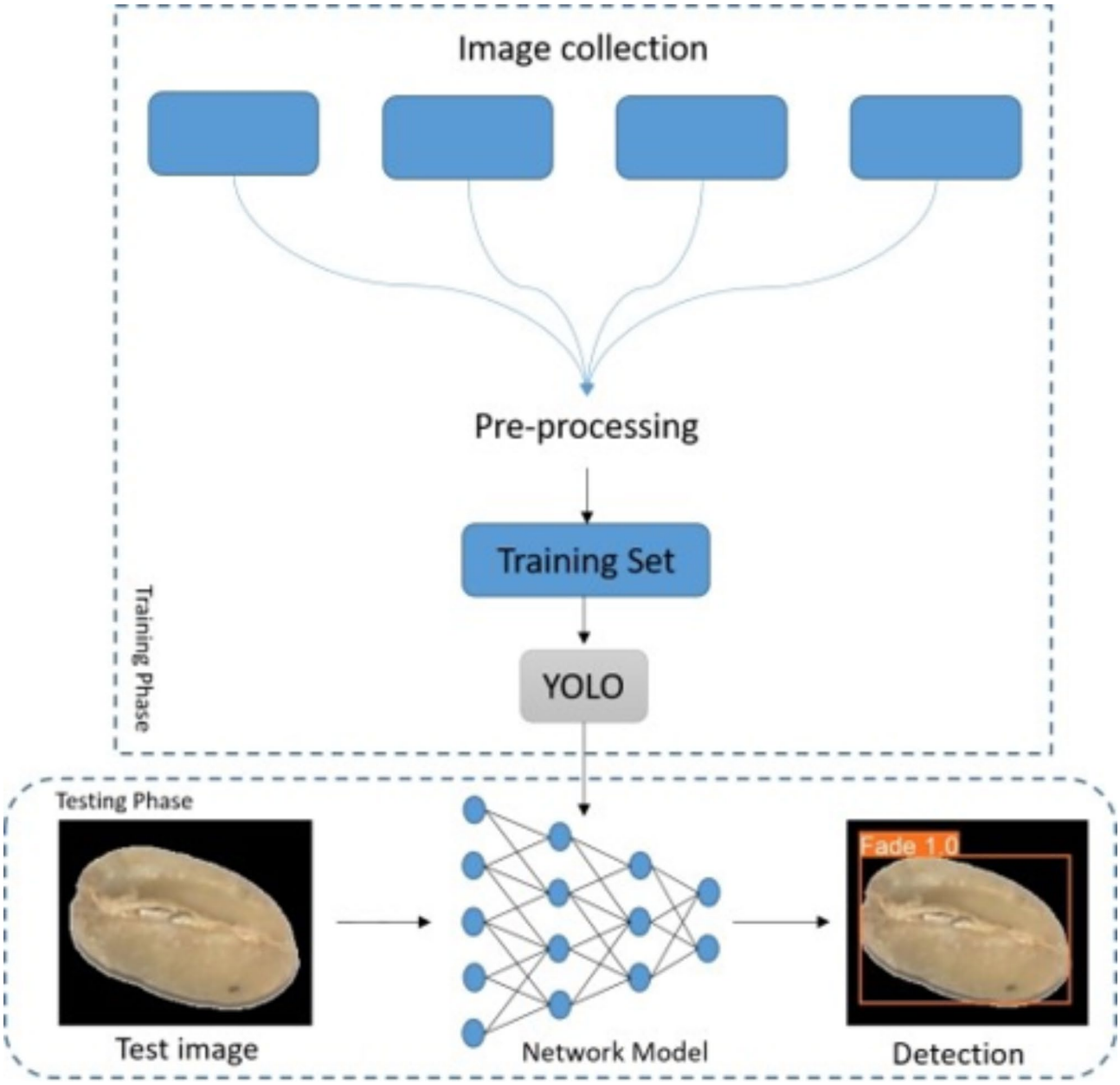

Fig. 6. Block diagram illustrating the full training process of the YOLO models for coffee bean defect detection. The process involves dataset preparation, training, and evaluation. The models are trained using the training set and evaluated on the test set using metrics such as precision, recall, and f1-score.

Configuration	Value
Optimizer	SGD
Epochs	150
Batch size	16
Learning rate	0.01
Momentum	0.937
Weight decay	0.0005, 0.001
Activation function	LeakyReLU, Mish, Sigmoid
Loss function	GIoU loss

Table 2. Hyper-parameter configuration.

recall, and f1-score were computed using Eqs. (1)-(3), which evaluate the model's performance on the training and test sets[23].

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$F1 - score = \frac{2 * (Precision.Recall)}{Precision + Recall} = \frac{2TP}{2TP + FP + FN} \tag{3}$$

Results and discussion

The study evaluated the effectiveness of six object detection models, including YOLOv3, YOLOv4, YOLOv5, YOLOv7, YOLOv8, and a custom model, on coffee bean categorization. The accuracy and recall values of each model were evaluated using a dataset of 5044 pictures of coffee beans divided into four groups. The YOLOv8 model and the custom-YOLOv8n model emerged as the top performers, demonstrating the effectiveness of customized models and cutting-edge architectures in object detection tasks, especially when trained on specific datasets with unique properties. The study emphasizes the importance of further research into model performance, including examining misclassifications and false positives/negatives to clarify the fundamental issues faced by each model. The YOLO models performed very well with 150 epochs. A suitable image resolution of 640 × 640 pixels was chosen in this study. To verify the effectiveness of the custom-YOLOv8n method, the coffee beans were tested in 506 test set images, and the results are shown in Table 3; Fig. 7.

Comparison of the YOLO model variants, including specific numerical differences in key performance metrics such as precision, recall, f1-score, and mean Average Precision (mAP). The included bar chart illustrates the performance of the different models, highlighting the numerical differences in their effectiveness. For instance, the performance of YOLOv3 yielded a precision of 0.741, recall of 0.680, f1-score of 0.709 and mAP of 0.770, while YOLOv8 achieved a precision of 0.959, recall of 0.944, f1-score of 0.946, and mAP of 0.952. Notably, our custom-YOLOv8n model performed the best, with a precision of 0.977, recall of 0.990, f1-score of 0.983, and mAP of 0.995.

The superior performance of the custom-YOLOv8n model can be attributed to several key modifications we implemented. First, we optimized the backbone of the model by adjusting the depth and width to better capture features from the dataset. Second, we modified the neck component, using a refined Feature Pyramid Network (FPN) to improve multi-scale feature fusion. Third, we tuned the anchor boxes to better align with the size and aspect ratios of the coffee beans and their defects, which further enhanced detection accuracy. Finally, we conducted extensive hyperparameters tuning, including adjustments to the learning rate, optimizer and other training parameters, to optimize detection performance, especially for small defects in the green coffee beans. These modifications collectively led to a marked improvement in the custom-YOLOv8n model, significantly boosting its precision and mAP.

From Fig. 8 two bars are plotted side by side for each class; the orange bar represents YOLOv8 and the green bar represents custom-YOLOv8n. It's clear from the Fig. 8 that both models perform almost identically for among the four classes, Custom-YOLOv8n has the highest accuracy performance. It should be acknowledged, nonetheless, that custom-YOLOv8n performs well in terms of accuracy. This shows that when it comes to recognizing and categorizing things inside these categories, both models are equally strong and trustworthy.

This confusion matrix, likely stemming from a custom-YOLOv8n experiment on employee image detection, offers valuable insights into the model's performance. Each cell visualizes how well objects from specific departments (represented by rows) were classified, with correct predictions highlighted on the diagonal. Deviations from the diagonal indicate misclassifications, allowing us to pinpoint classes the model struggles with. By calculating metrics like accuracy, precision, recall, and f1-score for each department (as shown in Fig. 9).

The method shows that recall and precision are positively correlated, with more recall translating into greater accuracy. At a confidence level of 0.5, the mean Average Precision (mAP) stays high for all classes, suggesting correct modeling for 0.995 of detection as shown in Fig. 10. However, during training and testing, the algorithm encountered difficulties because of a lack of representative data, which led to less-than-ideal performance at first. Notwithstanding these early challenges, the algorithm's rising accuracy in locating pertinent items indicates its performance in the task. The first challenges emphasize how crucial it is to have representative and varied training data for the best results.

Models	Input size	Precision	Recall	F1-score	mAP
YOLOv3	640 × 6 40	0.741	0.680	0.709	0.707
YOLOv4	640 × 6 40	0.665	0.631	0.648	0.628
YOLOv5	640 × 6 40	0.863	0.701	0.694	0.674
YOLOv7	640 × 6 40	0.775	0.777	0.776	0.826
YOLOv8	640 × 6 40	0.959	0.944	0.946	0.952
Custom-YOLOv8n	640 × 6 40	**0.977**	**0.990**	**0.983**	**0.995**

Table 3. Comparison of defection performance for several models.

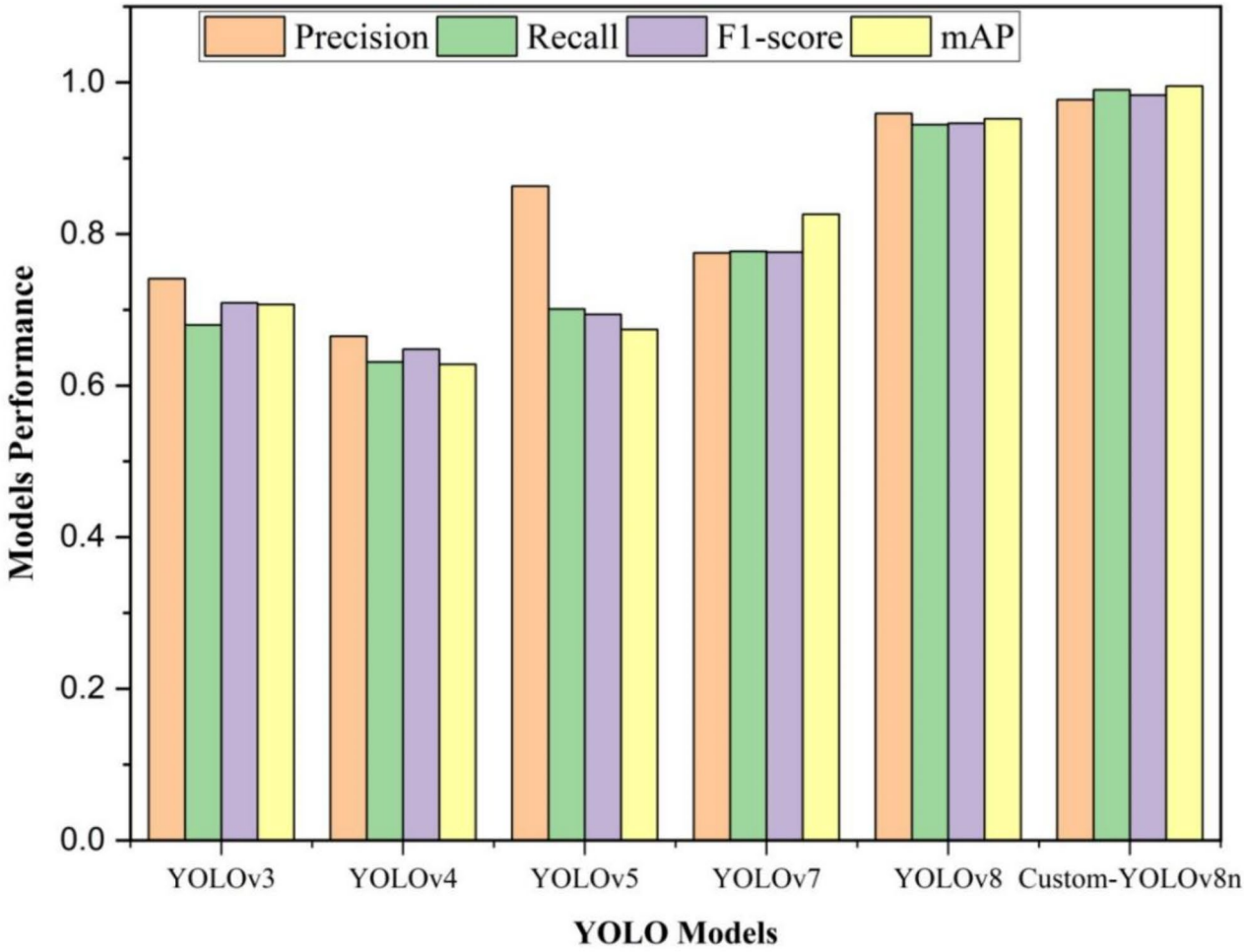

Fig. 7. The figure shows the performance comparison among the custom-YOLOv8n, YOLOv8, YOLOv7, YOLOv5, YOLOv4, and YOLOv3. The custom-YOLOv8n outperformed the YOLOv8, YOLOv7, YOLOv5, YOLOv4, and YOLOv3 in defective coffee bean detection.

The original dataset and the custom model were used to create the model for bean classification. The effectiveness of the model is illustrated by graphs that display various metrics from the training and validation sets. There are three different kinds of loss shown: categorization loss, objectness loss, and box loss (as shown in Fig. 11). The accuracy with which an algorithm can determine the center of an item and its bounding box is measured by its box loss. "Objectness" measures how likely it is that an object will be discovered at a specific location. Classification loss is a measure of how well an object's class was identified. As the number of iterations rises from 100 to 200, the model's performance is continually optimized.

Figure 12 represents some of the output images from trained model inference. The proposed custom-YOLOv8n model accurately detected the coffee bean defects on the coffee bean surfaces. The proposed model's ability to handle small objects and its improved performance with more training data highlight its potential for accurate and reliable defect detection on coffee bean surfaces. Compared to traditional methods, the proposed custom-YOLOv8n model seems to have provided better results in identifying these defects. In this study, the detection results showed that the proposed custom-YOLOv8n model could detect them with high confidence coefficients.

Conclusions and future works

Detecting defective beans is vital for improving coffee production processes, especially in developing countries like Bangladesh, where it could significantly boost the economy and improve the livelihoods of farmers. In this study, various YOLO models, including a customized YOLOv8 model, were employed to detect and classify four types of defective green coffee beans. The custom-YOLOv8n model achieved the best performance, with a recall of 0.990, precision of 0.977, and an f1-score of 0.983. This model outperformed all other YOLO variants in this study, demonstrating its effectiveness in automating quality control for the coffee industry.

The proposed custom-YOLOv8n model exhibited superior performance across all key metrics when compared to previous versions, suggesting its strong potential for practical applications in coffee bean defect detection. However, despite these positive results, the model's training process is still limited by long training

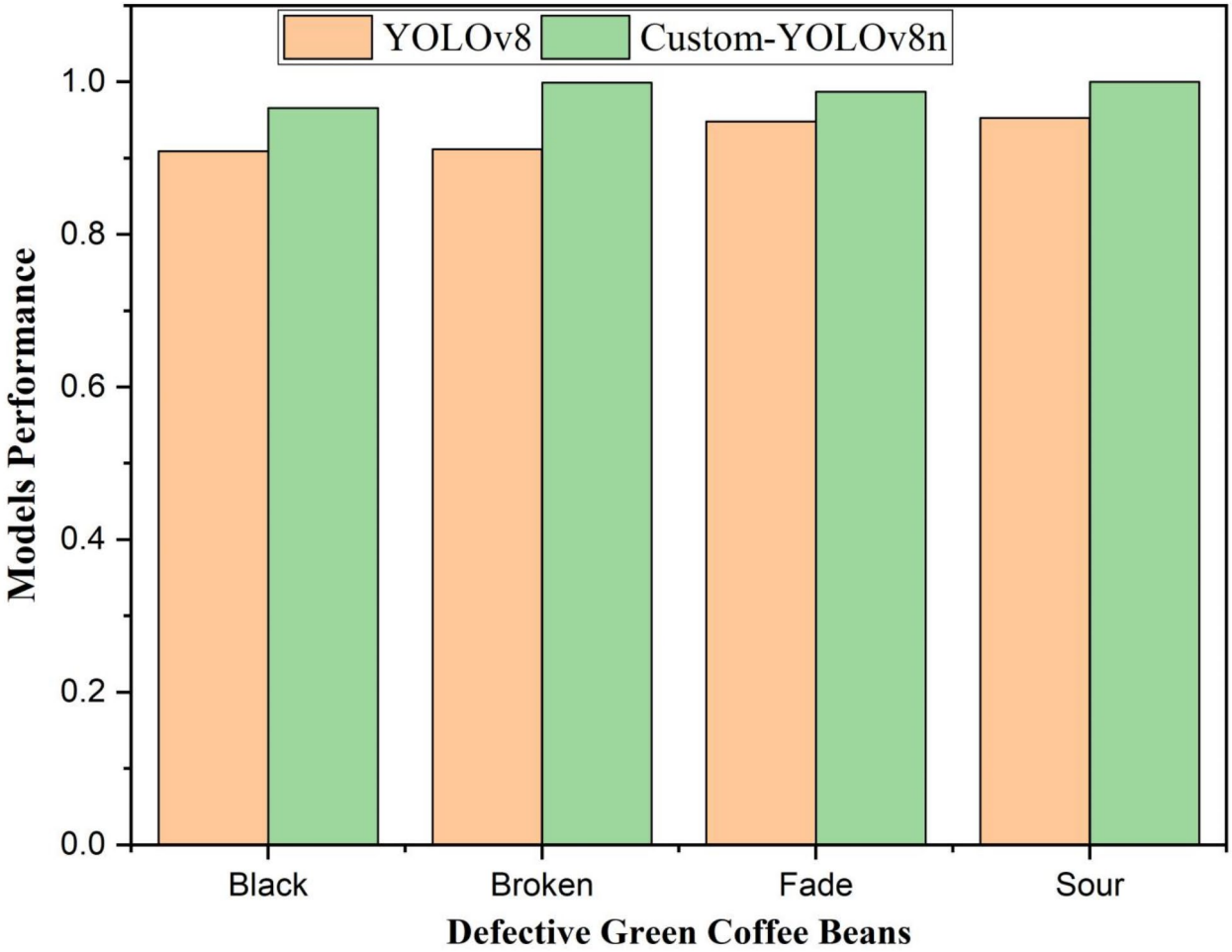

Fig. 8. Class-wise performance comparison between the two best performing models (YOLOv8 and custom-YOLOv8n). For every class, custom-YOLOv8n outperforms YOLOv8 with a narrow margin.

times. For future research, batch normalization techniques could be explored to further optimize training speed and accuracy. Increasing the dataset size by gathering a more diverse range of coffee beans from various types and fertility stages could also lead to improved generalizability. Additionally, advanced tagging methods could further enhance image quality and detection accuracy.

Future studies should also consider the integration of multi-modal data sources, including temperature, humidity, and soil conditions, to develop an early warning system for coffee bean defects based on multi-sensor fusion. This would provide a more comprehensive solution that could be adapted to real-world scenarios, including compatibility with Internet of Things (IoT) devices for real-time monitoring. Finally, the framework developed in this study could be easily modified to detect other plant species and crop diseases, potentially being deployed via mobile applications to assist farmers in real-time. Overall, this research contributes significantly to the automation of agricultural processes, offering practical tools for early and efficient coffee bean defect detection, which could lead to better crop management and production.

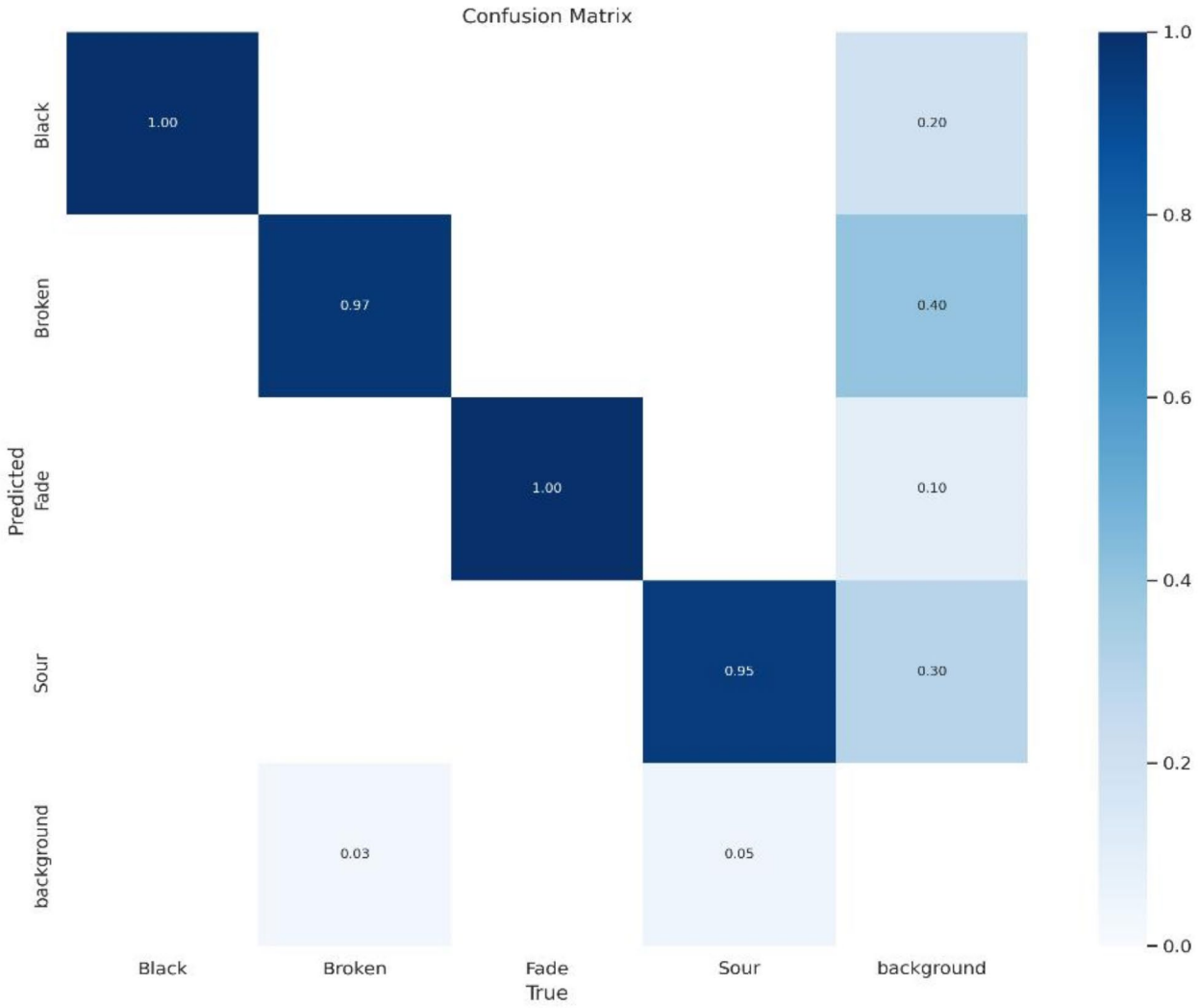

Fig. 9. Confusion Matrix of custom-YOLOv8n model.

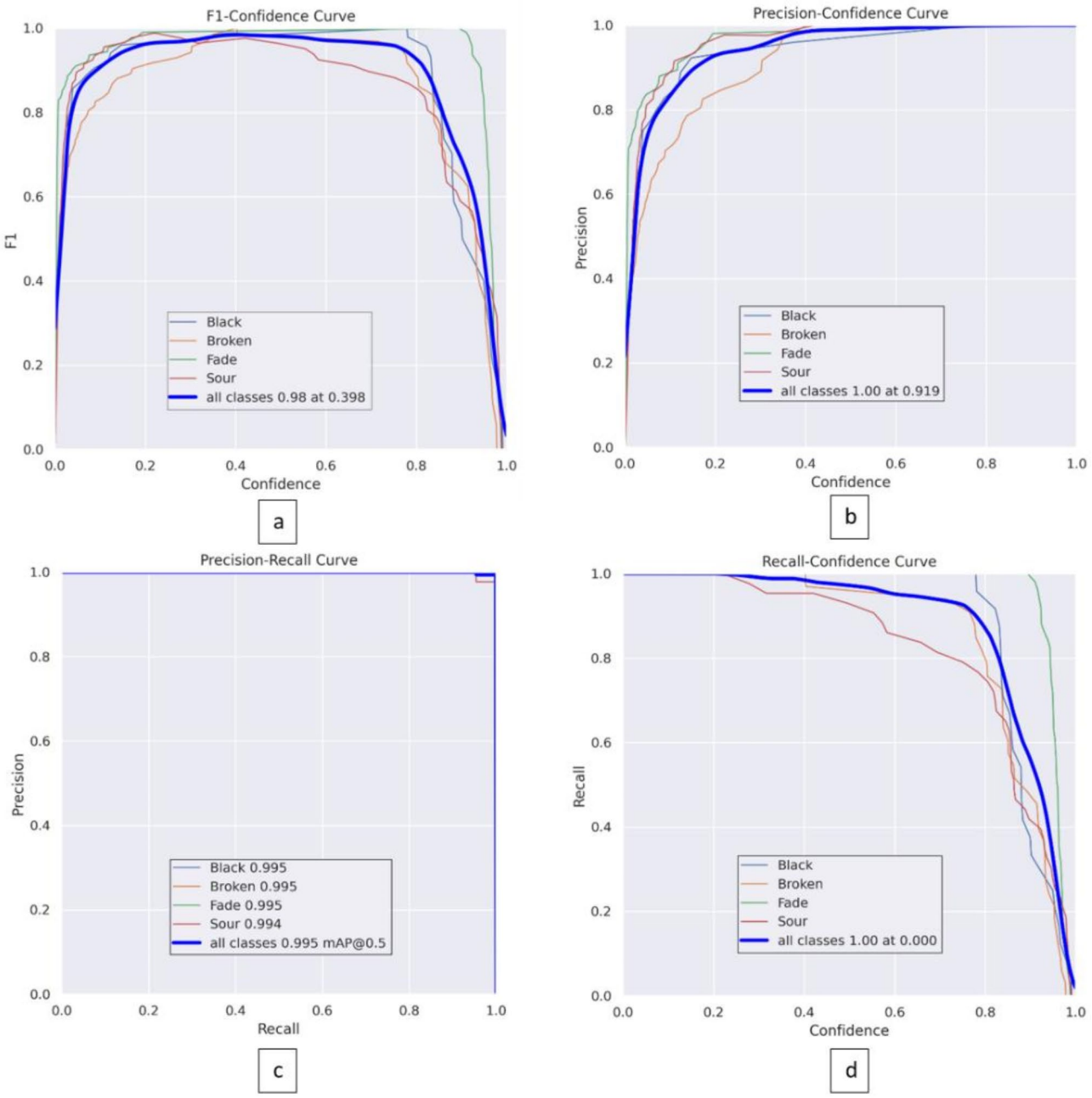

Fig. 10. Operation results curve; (**a**) f1-confidence curve, (**b**) precision-confidence curve, (**c**) precision-recall curve, and (**d**) recall-confidence curve.

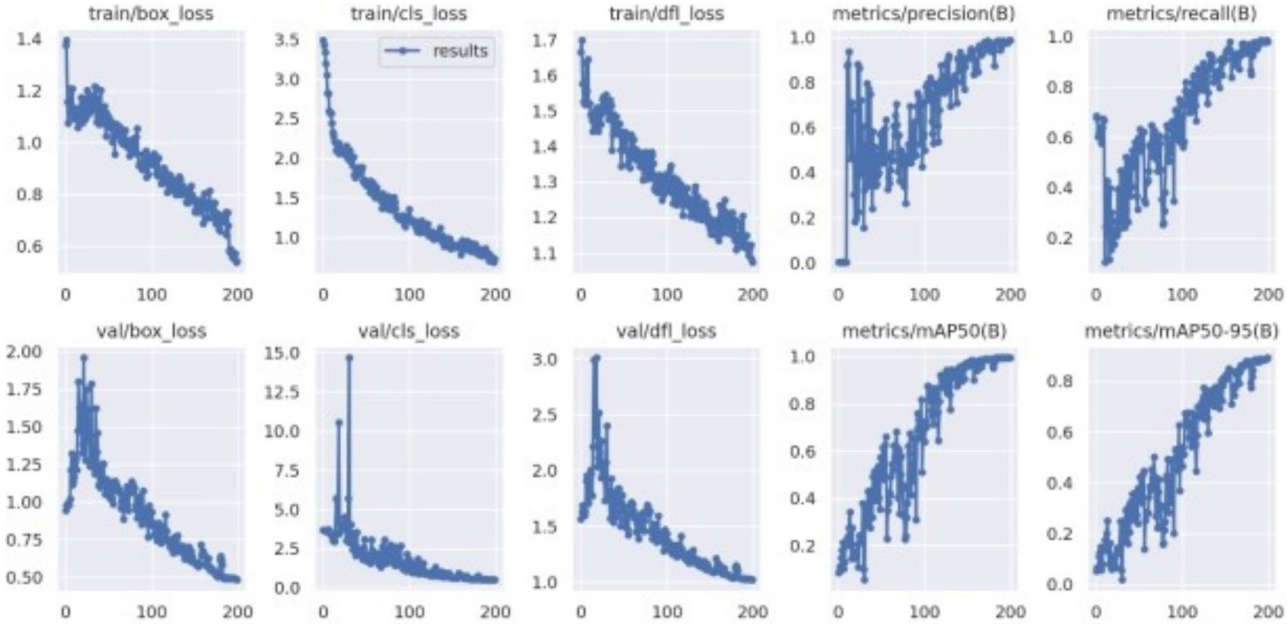

Fig. 11. Visual analysis of model evaluation indicators (precision, recall, and mAP@0.5 for the proposed model) during training.

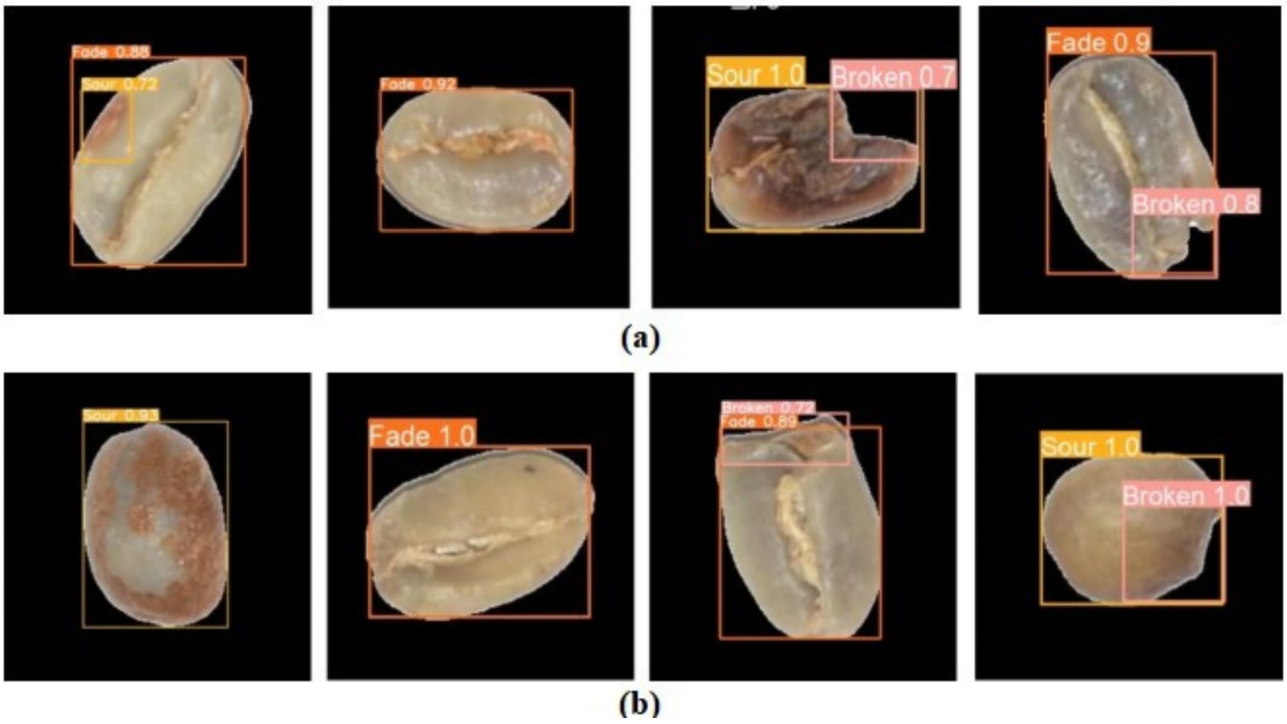

Fig. 12. The figure (**a**) and (**b**) represent the visual demonstration of the detection results of the proposed custom-YOLOv8n.

Data availability

All the data used in this study can be obtained from the corresponding author upon a reasonable request. Note: If anyone needs to request the data, feel free to reach out to me (hlgope@sau.ac.bd).

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Author contributions

The conceptualization, programming, writing, reviewing, and finalization of the manuscript were led by Hira Lal Gope and Hidekazu Fukai, while Hira Lal Gope and Fahim Mahafuz Ruhad were responsible for coding, as well as training and testing the model. Hidekazu Fukai contributed significantly to data collection and review writing. Shohag Barman aided in writing, reviewing, and editing. Both Hira Lal Gope and Fahim Mahafuz Ruhad played vital roles in data processing, writing, and reviewing, ensuring the manuscript's completion. All authors reviewed and stand accountable for the work, committed to addressing questions on its accuracy.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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